Project Summary

Batch details	
Team members	05
Domain of Project	
Proposed project title	Airfare Price Prediction Using Machine Learning
Group Number	09
Team Leader	Sachin Anthony
Mentor Name	Aishwarya Sarda

Date:

Signature of the Mentor

Signature of the Team Leader

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Project Details

OVERVIEW:

Airfare prices in the U.S. vary significantly across different airport routes due to multiple factors such as flight distance, passenger demand, airline competition, market size, and seasonal travel patterns. Understanding these variations is essential for airlines, policymakers, and travelers who rely on accurate pricing insights for planning and decision-making. This project uses the Consumer Airfare Report dataset, which contains detailed information about U.S. airport-pair markets, to analyze pricing behavior and identify key factors that influence airfare. By applying exploratory data analysis and machine learning techniques, the project aims to develop a predictive model capable of estimating airfare for any given route and generating valuable insights into the dynamics of airline pricing.

Business problem statement (GOALS)

1. Business Problem Understanding

   Airfare pricing is a complex and dynamic process. Airlines change fares based on operational costs, route distance, competition, seasonal trends, passenger demand, and revenue-management algorithms. These pricing models are proprietary and not publicly available.

   This lack of transparency causes challenges:

   - Travelers cannot estimate whether ticket prices are fair or overpriced.
   - Travel agencies struggle with route planning and price forecasting.
   - Airlines cannot easily compare competitiveness across markets.
   - Policymakers lack data to analyze market fairness

2. Business Objective

   The primary business objective of this project is to predict the average airfare price for any U.S. airport route and identify the major factors influencing these prices. Since airfare frequently fluctuates due to factors such as distance, passenger demand, seasonal trends, and airline competition, it becomes challenging for stakeholders—including airlines, travel companies, and passengers—to make informed decisions.

   This project utilizes the Consumer Airfare Report dataset to develop a transparent, data-driven predictive system that helps users understand airfare behavior and supports better decision-making.

3. Approach
   i. Explore and clean the Consumer Airfare dataset
   ii. Analyze route characteristics and pricing behaviour
   iii. Engineer meaningful features such as fare-per-mile and seasonal trends
   iv. Build machine learning regression models to predict airfare
   v. Interpret key factors influencing fare variations

4. Conclusions

   The project aims to deliver a predictive system capable of estimating airfare for any U.S. route and provide data-driven insights into pricing behavior, helping stakeholders make optimized business decisions.

TOPIC SURVEY

1. Problem understanding

   Airfare prices change often because of distance, demand, season, airline competition, and other travel costs. These frequent changes make it hard for people to understand why prices go up or down. Airlines use advanced pricing systems to decide fares, but these systems are very expensive, difficult to maintain, and not available to the public.
   Because of this, passengers cannot plan their travel properly, travel companies find it hard to give accurate price suggestions, and analysts cannot judge if prices are fair. The high cost of using and maintaining airline pricing systems also creates a big problem.
   To solve this, we need a simple, low-cost, data-driven model that studies the factors affecting prices and predicts airfare clearly. This will help all users make better decisions.

2. Current solution to the problem

   Existing pricing systems primarily rely on airline revenue management algorithms, historical fare data, and competitive positioning. However, these systems are proprietary and not openly shared. Public research typically focuses on descriptive analysis of airfare trends, with fewer studies applying machine learning to forecast fare values using large-scale route-level datasets.

3. Proposed solution to the problem

   This project proposes a predictive modeling approach using machine learning to estimate airfare prices based on route characteristics. By leveraging the U.S. Consumer Airfare Report dataset, the project identifies determinants such as distance, quarterly seasonality, passenger volume, airport pair, and carrier influence. Regression models such as Linear Regression, Random Forest, and Gradient Boosting will be applied to generate accurate predictions.

4. Reference to the problem

   Real-world studies, airline market reports, and government aviation statistics highlight substantial variation in airfare patterns. Publicly available datasets (e.g., BTS Consumer Airfare Report) provide rich information useful for building predictive models and understanding pricing economics. This project extends that foundation using ML techniques

CRITICAL ASSESSMENT OF TOPIC SURVEY

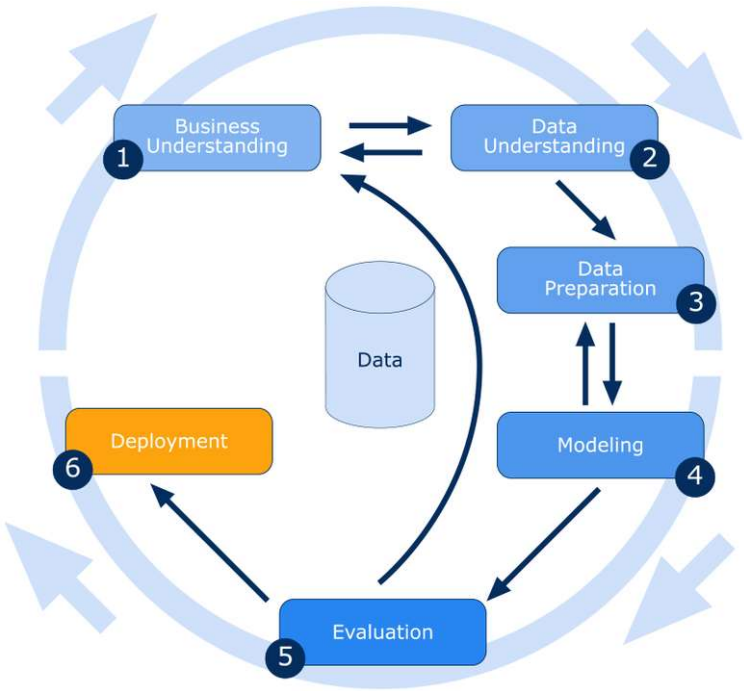
1. Find the key area, gaps identified in the topic survey where the project can add value to the customers and business?

   The topic survey reveals that while airfare data is publicly available, most analyses focus on historical trends rather than providing forward-looking insights. Airlines often use internal, proprietary pricing systems, leaving customers and policymakers with limited visibility into why fares change across routes. This project adds value by creating a transparent, data-driven prediction model that helps customers anticipate price patterns and enables businesses, especially airlines and travel agencies, to make informed decisions on pricing strategies, demand forecasting, and competitive analysis.

2. What key gaps are you trying to solve?

   The major gaps identified include the lack of accessible predictive models for airfare estimation, limited research applying machine learning to route-level airfare data, and an absence of clear explanations for the factors influencing fare variations. This project addresses these gaps by developing an interpretable machine learning framework that predicts airfare and highlights the most significant drivers of pricing. The goal is to bridge the knowledge gap and provide actionable insights for users who currently rely on guesswork or non-transparent fare systems.

METHODOLOGY TO BE FOLLOWED



1. **Business Understanding**

   The objective of this project is to predict airfare prices for U.S. airport routes and understand the factors that drive pricing variations. From a business point of view, this project supports three major groups. **Airlines** can use the insights to study route performance and improve their pricing decisions. **Travel platforms and agencies** gain a clearer understanding of market patterns, helping them guide customers more accurately. **Policymakers** can use the findings to review whether pricing practices are fair across different markets.

2. **Data Understanding**

   The dataset used in this project is the **U.S. Consumer Airfare Report**, which provides detailed route-level airfare information across the United States.

   During data understanding, the dataset was checked for:

   - **Shape of the data** (255,780 rows × 23 columns)
   - **Variable types** (numerical, categorical, identifiers)

     Numerical variables:

     Year, quarter

     nsmiles (route distance)

     passengers (number of travelers)

     fare (average airfare)

large_ms (market share of the largest carrier)

fare_lg, lf_ms, fare_low (additional fare and market share information)

Categorical variables:

Year, quarter

city1, city2

airport_1, airport_2

carrier_lg, carrier_low

Geocoded_City1, Geocoded_City2

Identifier variables:

citymarketid_1, citymarketid_2

airportid_1, airportid_2

tbl, tbl1apk

- **Summary statistics:**

	count	mean	std	min	25%	50%	75%	max
Year	255780.0	2009.133904	9.064018	1993.0000	2001.0000	2009.0000	2017.0000	2025.0
quarter	255780.0	2.476374	1.118472	1.0000	1.0000	2.0000	3.0000	4.0
citymarketid_1	255780.0	31556.026456	1090.557917	30135.0000	30721.0000	31423.0000	32467.0000	35412.0
citymarketid_2	255780.0	32180.764419	1233.742285	30189.0000	30994.0000	32211.0000	33192.0000	35628.0
airportid_1	255780.0	12436.084612	1431.049035	10135.0000	11193.0000	12266.0000	13487.0000	16440.0
airportid_2	255780.0	13247.691239	1425.643614	10466.0000	12197.0000	13303.0000	14679.0000	15919.0
nsmiles	255780.0	1189.721331	702.463779	109.0000	629.0000	1023.0000	1735.0000	2724.0
passengers	255780.0	304.323153	518.037063	0.0000	22.0000	115.0000	345.0000	8301.0
fare	255780.0	220.311466	82.382791	50.0000	165.8300	210.8200	264.4900	3377.0
large_ms	254222.0	0.667205	0.224663	0.0038	0.4852	0.6593	0.8800	1.0
fare_lg	254222.0	220.078323	84.864548	50.0000	162.6800	209.8200	265.3875	2725.6
lf_ms	254150.0	0.453176	0.333487	0.0100	0.1600	0.3600	0.7600	1.0
fare_low	254150.0	191.494950	73.986645	50.0000	140.1825	182.5600	231.6500	2725.6

3. **Data Preparation**

The Data Preparation stage ensures that the dataset is clean, consistent, and ready for analysis and model building. The following steps were performed:

1. **Handling Missing Values**:

   The dataset contained missing values in several columns such as carrier_lg, large_ms, fare_lg, carrier_low, lf_ms, fare_low, and geocoded location fields.

- **Mean/Median** for numerical columns.
- **Mode** for categorical variables
- Columns like Geocoded_City1 and Geocoded_City2 had extremely high null counts, so they were **dropped** to avoid noise and reduce dimensionality.

2. **Detecting and Treating Outliers:**

   Outliers create skewed distributions and affect model performance.

   Steps Performed

- Used **boxplots** and **IQR method** for outlier detection.
- Outliers were treated by:

  Capping extreme values at upper/lower bounds

3. **Encoding Categorical Variables:**

   Machines learning models only accept numerical values.

   Categorical variables such as **airport codes, carriers, city identifiers** must be converted to numeric format.

   Why Encoding is Required?

   - Algorithms cannot understand text like "ATL", "JFK", or "DL"
   - Encoding preserves the information in numerical form
   - Helps models learn patterns without interpreting categories incorrectly

   Methods Used

   - **Label Encoding** for ordinal or high-cardinality categorical variables
   - **One-Hot Encoding** for nominal variables such as airport IDs

4. **Exploratory Graph Analysis:**

   To understand variable distribution and relationships:

   a) **Distribution of Continuous Variables (Univariate Analysis):**

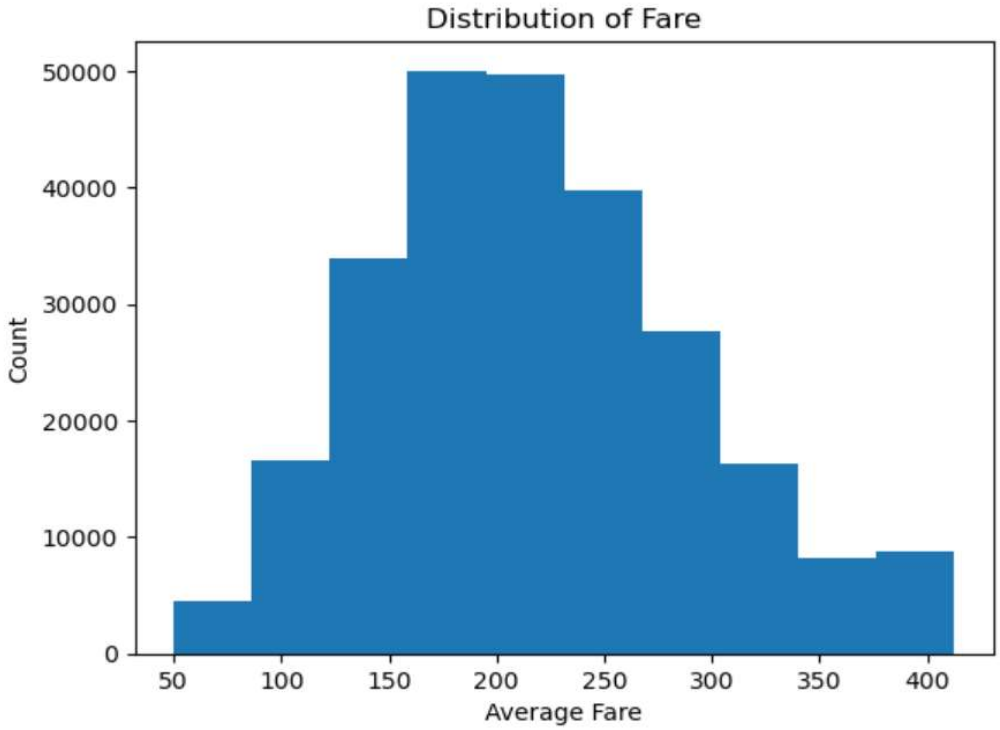
      Plotted histograms (show distribution), and

      boxplots (show outliers + spread)

      for (Fare, passengers, nsmiles)

   Purpose:

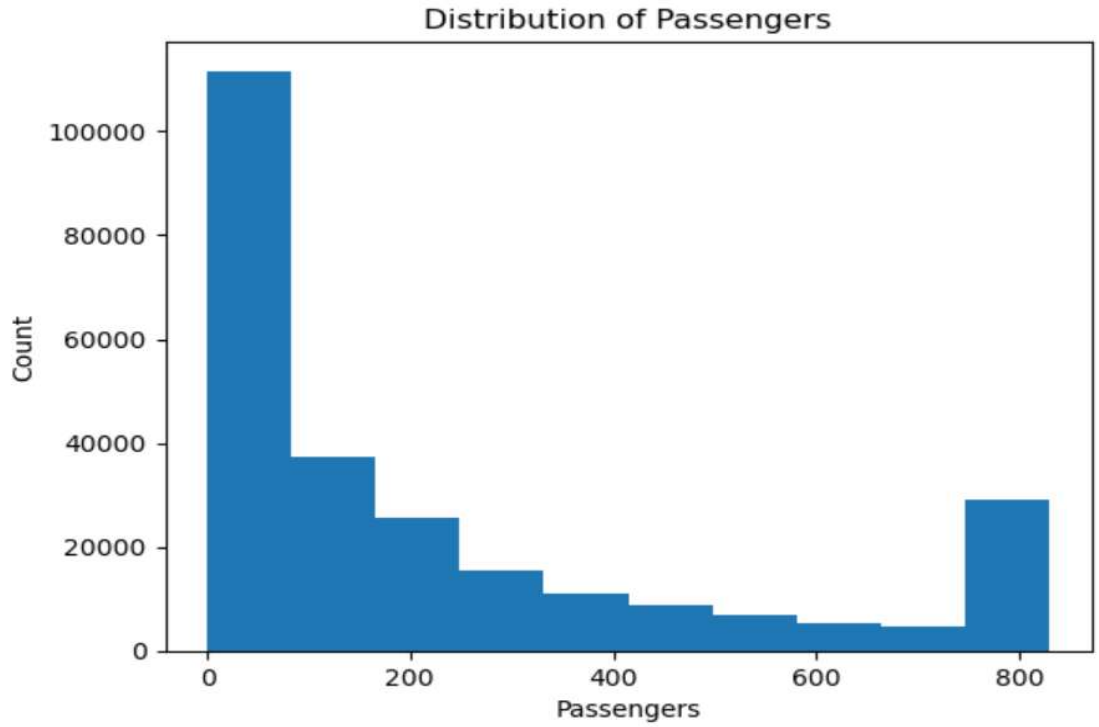
   To observe normality, skewness, and outliers.



Purpose:

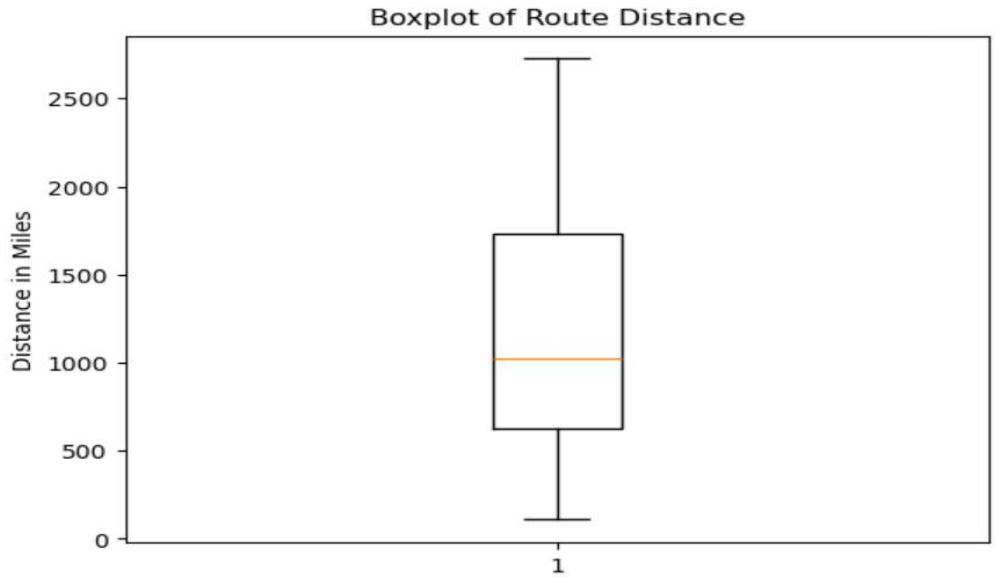
To see the distribution of fare and detect outliers so I can prepare the data properly for modeling.

It shows most fares fall between 150–250, very few are extremely high or low, and the distribution looks slightly spread with some outliers at the high end.



Purpose:

Count represents the number of records that fall within each passenger range.



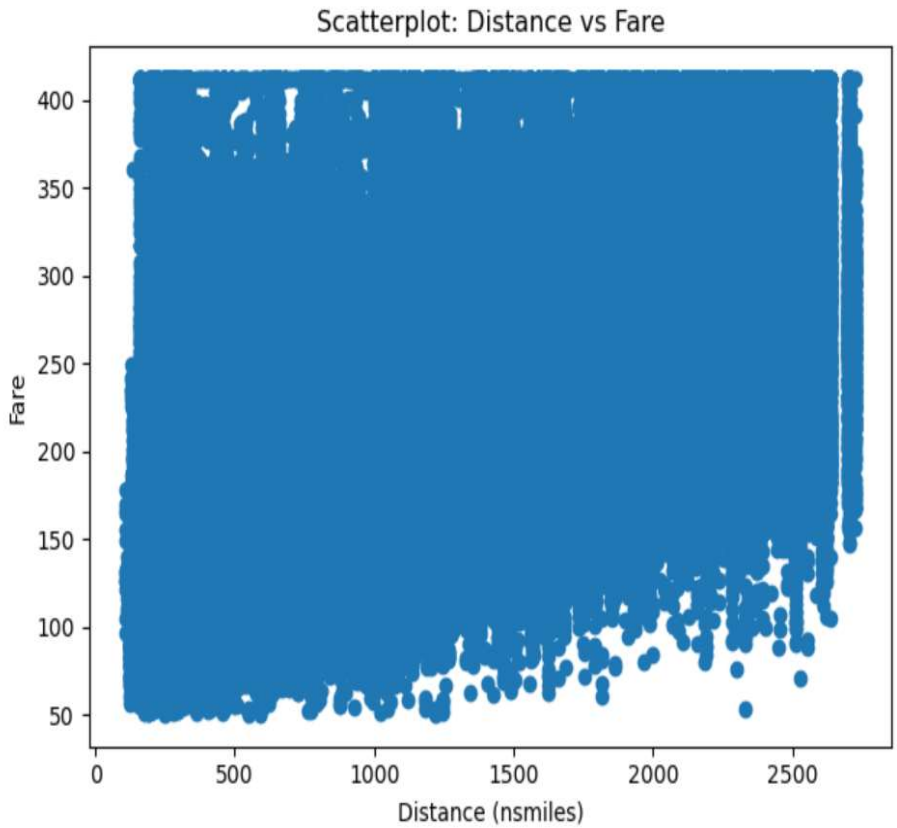
Purpose:

To detect outliers in route distance and understand how distances vary across U.S. flight routes.

b) **Scatterplots for Relationships (Bivariate Analysis):**

Created scatterplots between key numerical columns such as:

- **Nsmiles(distance) vs fare**



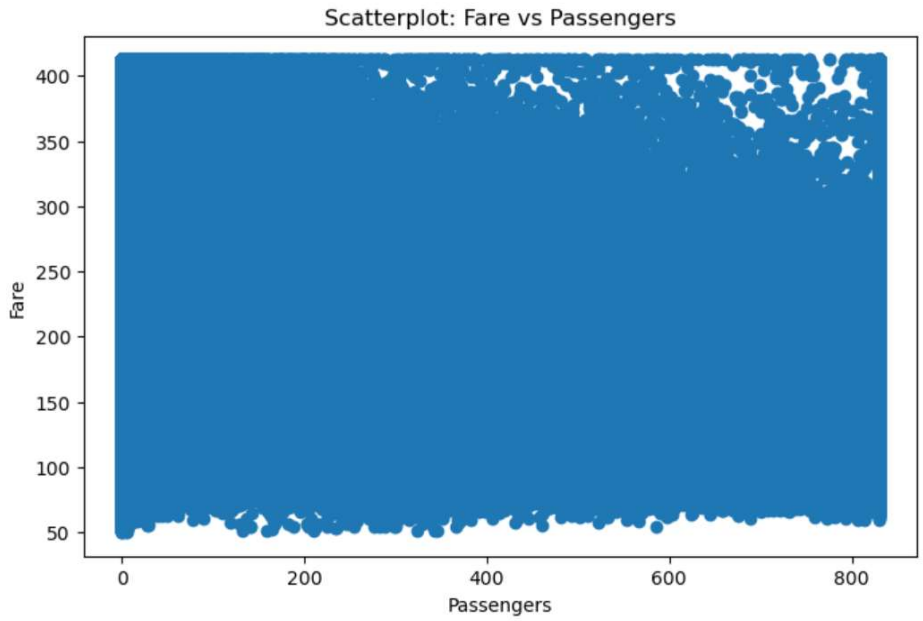
Purpose:

To understand how airfare changes with travel distance.

This helps identify patterns such as:

- whether longer routes have higher fares
- whether the relationship is linear or nonlinear
- presence of cost anomalies for certain distances

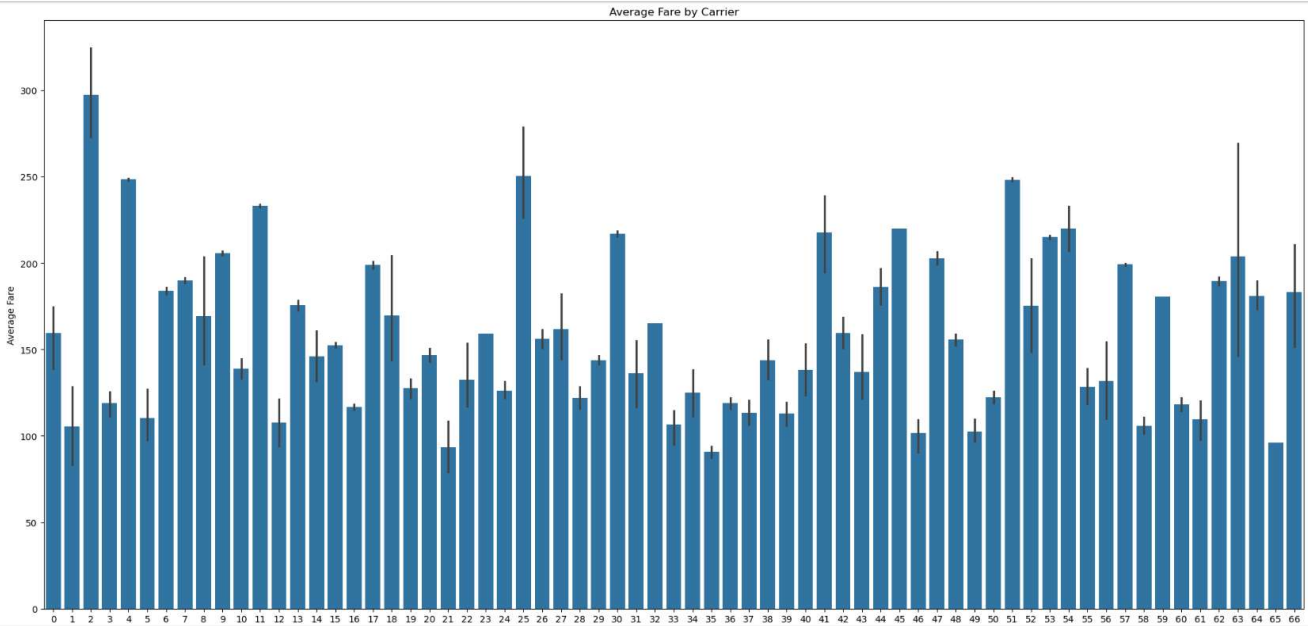
- **fare vs passengers:**



Purpose:

- Shows how fare changes with the number of passengers
- Helps identify trends or patterns

c) carrier_lg vs fare:



Purpose:

Shows which carriers are expensive vs. affordable

- Airlines with very tall bars → premium or expensive carriers
- Airlines with short bars → budget or lower-priced carriers

4. **Modeling**

   Regression models including Linear Regression, Random Forest Regressor, XGBoost, and Gradient Boosting are applied. Hyperparameter tuning and cross-validation ensure optimal performance. Model evaluation metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 score.

5. **Evaluation**

   Models are evaluated using standard performance metrics to ensure accuracy and reliability.

   Key success metrics used in this project include:

   - **MAE (Mean Absolute Error):** Measures how far predicted fares are from actual fares on average. Lower MAE indicates better accuracy.
   - **RMSE (Root Mean Squared Error):** Penalizes larger errors more heavily, helping identify models that handle price variability well.
   - **R^2 Score (Coefficient of Determination):** Shows how well the model explains variance in airfare prices. Higher values indicate stronger predictive power.
   - **MAPE (Mean Absolute Percentage Error):** Helps understand prediction error in percentage form, making results easy to interpret.

   Models are interpreted to identify the most influential features and analyze route-specific pricing patterns. Results are validated using test data to ensure the predictions are robust, generalizable, and suitable for real-world use.

REFERENCES

a. Airline Fare Data Source

U.S. Department of Transportation, Bureau of Transportation Statistics. *Domestic Airline Consumer Airfare Report (Table 1A)*.

https://www.bts.gov/air-fares

b. Industry Fare Standards
Standard Industry Fare Level (SIFL) Updates.
https://www.transportation.gov/

c. Airline Statistics
U.S. Department of Transportation, Office of Aviation Analysis.
U.S. International Air Passenger and Freight Statistics Report.
https://www.transportation.gov/policy/aviation-policy/us-international-air-passenger-and-freight-statistics-report

Research Paper:

d. Borenstein and Rose (1994) studied U.S. airline fares using the same DOT Origin–Destination dataset as this project. They showed that airfare mainly depends on factors like distance, competition, and the airline operating the route. This supports using features such as nsmiles, fare, carrier, airports, and city-pair markets in the model.
https://faculty.haas.berkeley.edu/borenste/download/JPE94AirPrice.pdf?utm_source=chatgpt.com

e. Wang et al. (2019) proposed a machine-learning framework for airfare price prediction using U.S. airline datasets such as DB1B and T-100, which contain key variables like fare, route distance, number of passengers, and carrier details. Their methodology supports the feature selection in this project, particularly the use of nsmiles, passenger count, carrier, and city-pair information for building an accurate airfare prediction model.
https://lweb.umkc.edu/chen/PDF/IRI2019_Wang.pdf?utm_source=chatgpt.com

f. Prasad et al. (2020) used machine learning methods (Random Forest, XGBoost, Linear Regression) to predict flight ticket prices. Their research shows that ML is an effective approach for airfare prediction, supporting the modeling techniques used in this project.https://www.researchgate.net/publication/382607145_Prediction_of_Airline_Ticket_Price_Using_Machine_Learning_Method

Original owner of data	Office of the Secretary of Transportation, U.S. Department of Transportation
Data set information	Consumer Airfare Report: Table 1a – All U.S. Airport Pair Markets. Provides quarterly data on average fares, passenger volumes, distances, and airport pairs. Aggregated as directionless markets.
Any past relevant articles using the dataset	SkyPredict: Airfare Prediction Model" (IOSR-JCE)
Reference	BTS Consumer Airfare Report – Table 1a
Link to web page	https://catalog.data.gov/dataset/consumer-airfare-report-table-1a-all-u-s-airport-pair-markets

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